

Apache CloudStack: A Cloud Computing Platform that Meets Your Needs

With its comprehensive feature set, flexible architecture, and strong community support, CloudStack is an attractive option for organisations looking to harness the full potential of cloud computing.



Image Source : <https://cloudstack.apache.org/>

Apache CloudStack is a comprehensive, open source cloud computing platform that provides a complete solution for creating, managing, and deploying scalable infrastructure as a service (IaaS) clouds. Developed by the Apache Software Foundation, CloudStack is designed to operate as an efficient and user-friendly cloud orchestration platform that can handle large networks of virtual machines. It is utilised by a variety of service providers to offer public cloud services, as well as by numerous businesses for on-premises private clouds or as part of a hybrid cloud solution.

History and evolution

The journey of CloudStack began with a company called Cloud.com, which was later acquired by Citrix Systems in 2011. Recognising the potential of

CloudStack to drive the future of cloud computing, Citrix contributed the project to the Apache Software Foundation. This move was aimed at leveraging the collaborative power of the open source community to further develop and enhance CloudStack. It became a top-level Apache project in 2013 and has since been nurtured by a diverse and active community of individual developers and corporate contributors.

Core features and functionalities

Apache CloudStack is packed with a multitude of features that cater to the essential requirements of cloud computing for both public and private cloud scenarios. Its robust capabilities include:

Compute orchestration:

CloudStack efficiently manages the deployment of virtual machines across

various compute resources, ensuring optimal utilisation of the infrastructure.

Network as a service (NaaS): It provides a suite of networking services, including load balancing, VPN setups, and firewall configurations, to maintain accessibility and security for cloud-hosted applications.

User and account management:

With comprehensive account management features, CloudStack offers granular access controls, enabling the creation and management of users, projects, and domains.

Resource accounting: The platform includes an integrated resource accounting system that tracks the consumption of cloud resources, which is crucial for billing and chargeback mechanisms.

API compatibility: CloudStack features an API that is compatible with AWS EC2 and S3, facilitating the